PROMPT ENGINEERING AND GENERATIVE AI

Modula, AI-assisted configuration of modular smartphones



The phone shown is the build Modula chose for a wedding photographer. The module in blue serves their top priority.

TEAM, ALPHABETICAL

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Problem, statement

A fixed phone gives everyone the same compromise. A modular phone moves every decision to the buyer, and the parts affect each other: a bigger battery adds hours and weight, and modules compete for space on the frame.

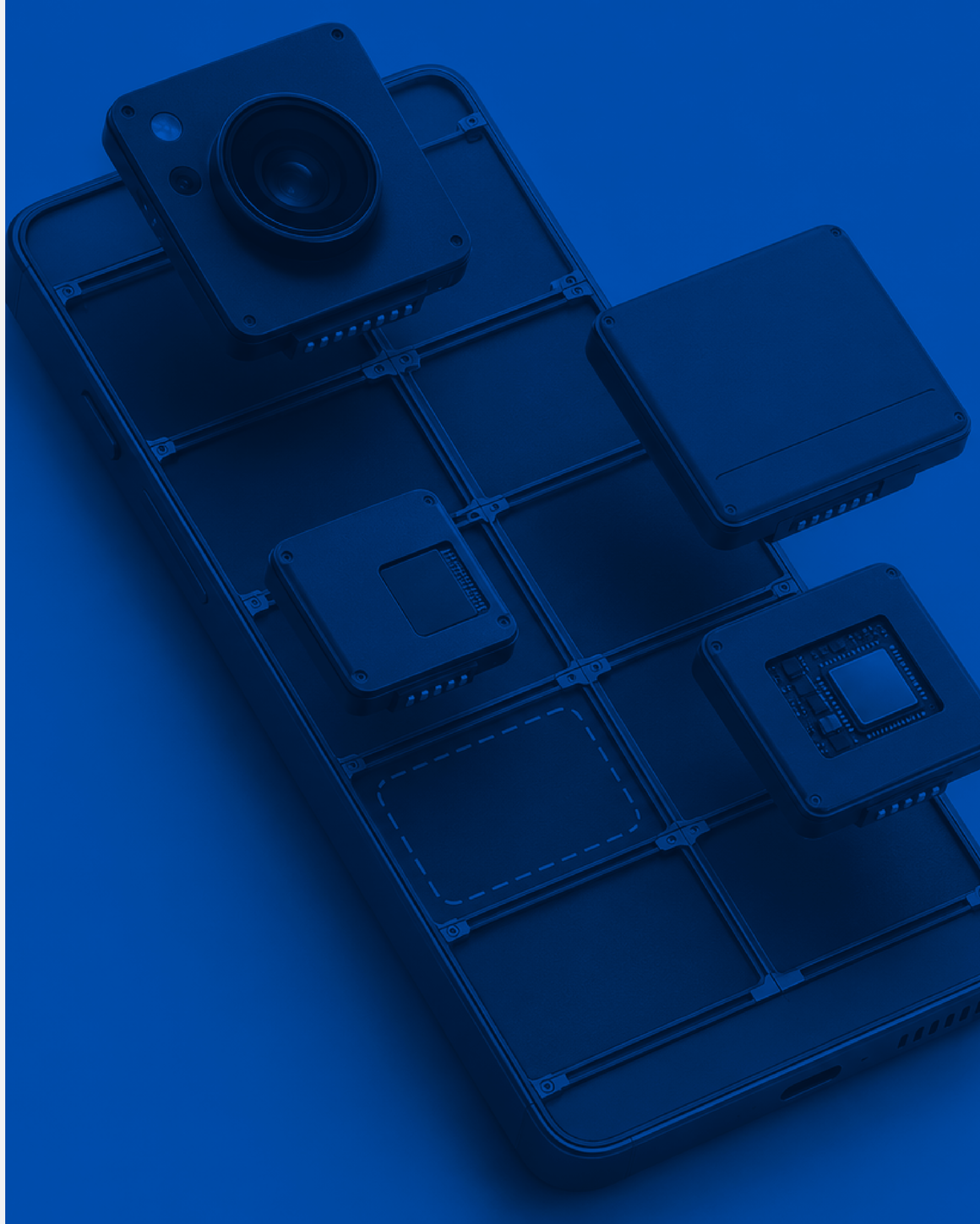
46,656 combinations

16,420 fit on the frame

1 best suits the person

Can Generative AI turn a plain-language description into a phone that fits the person and can physically be built?

23 modules in our sample catalogue. Google suspended Project Ara in 2016; reports cited cost, battery life and slower links between parts (Fortune, 2016).



The methodology

▪ CLAUDE ▪ PYTHON

1

Parse

Text to JSON: priorities, budget and required modules

2

Vector

Need and phones on the same nine axes

3

Search

Filter by space, budget and dependencies, then rank

4

Place

Fit modules on the frame by heat, antenna and grip

5

Explain

Two sentences written only from computed facts

PROMPTS COMPARED

- **v1** a minimal request
- **v2** role, axis definitions, anchored scale, strict schema
- **v3** v2 plus inference rules, reasoning step, few-shot examples

BUILD SCORE

0.5 × cosine + 0.5 × attainment − 0.2 × price ÷ ₹90,000

Cosine checks the phone's shape, attainment its strength on what matters, and price limits overkill.

The results

Tested once on 16 personas held back from tuning. Prompt v3 was the only prompt with no failed builds.

METHOD	CLEAN JSON	MODULES CAUGHT	MODULES CORRECT	BUDGET READ	FIT	FAILED BUILDS
One fixed phone	n/a	n/a	n/a	n/a	0.51	7
Keyword rules, no AI	n/a	100%	100%	100%	0.74	0
Prompt v1	19%	100%	21%	75%	0.64	3
Prompt v2	100%	13%	100%	100%	0.77	2
Prompt v3	**100%**	**100%**	**89%**	**100%**	**0.76**	**0**

PROMPT V3

1.5×

the fit of one fixed phone

0

failed builds across 16 held-out personas

16/16

personas fit better than the fixed phone

With v3, fit was 1.5 times that of the fixed phone and higher for all 16 personas. v2 scored 0.01 higher on fit, but caught only 13% of required modules.

Limits: labels are ours, catalogue figures are illustrative and the personas are written, not real users. The keyword list was written with these personas in view, which likely flatters it.

The references

MODULAR PHONES

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Tools. Claude, Python, NumPy, matplotlib and Streamlit.